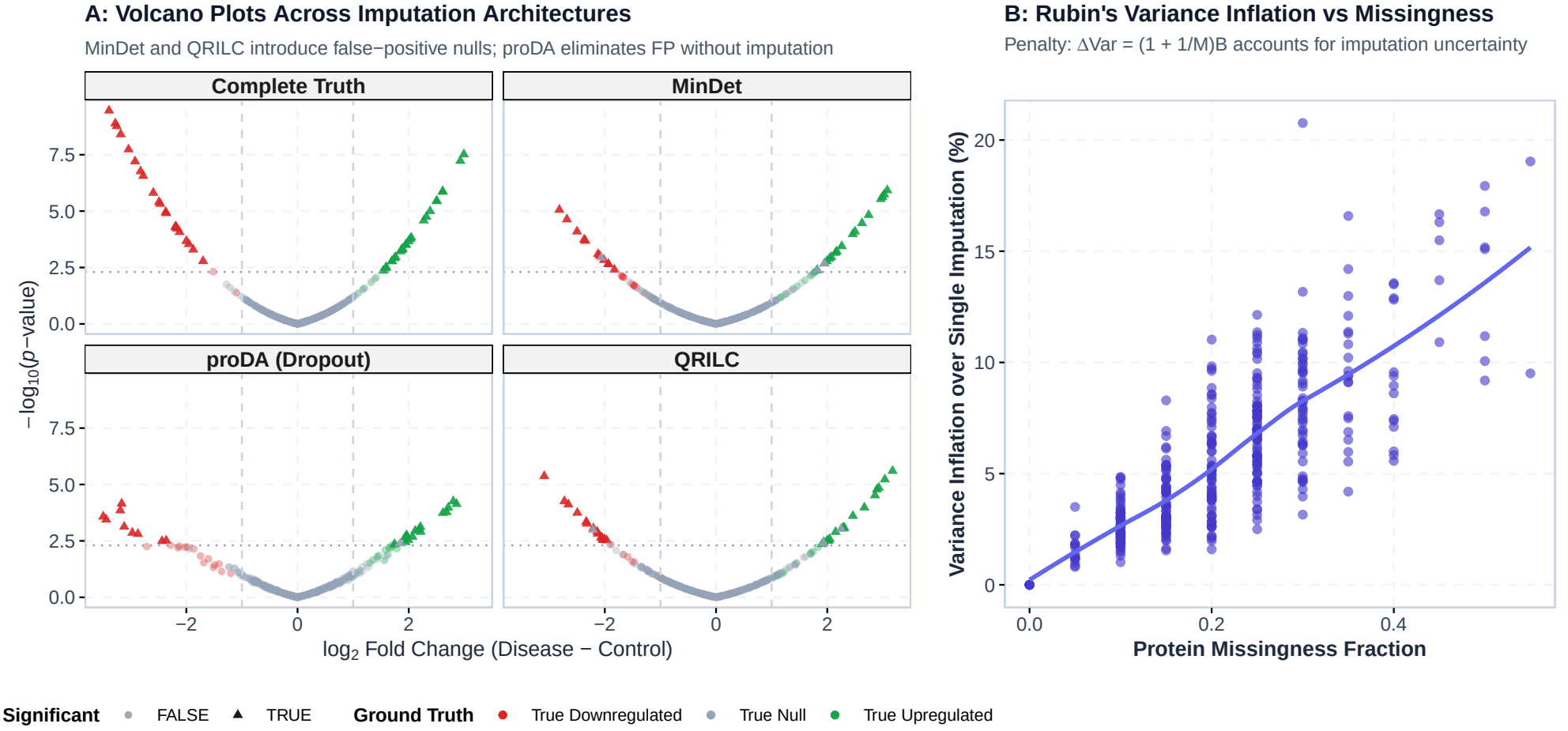

A: Volcano Plots Across Imputation Architectures
MinDet and QRILC introduce false−positive nulls; proDA eliminates FP without imputation
Complete Truth
MinDet
proDA (Dropout)
QRILC
−log10(p−value)
log2 Fold Change (Disease − Control)
B: Rubin's Variance Inflation vs Missingness
Penalty: ΔVar = (1 + 1/M)B accounts for imputation uncertainty
Variance Inflation over Single Imputation (%)
Protein Missingness Fraction
Significant
FALSE
TRUE
Ground Truth
True Downregulated
True Null
True Upregulated